

ACCTG 211H — Mechanics Drill Set (Answer Key)

Week 12 — Variances, ROI, Residual Income, and EVA

Drill Set

Required: 1–2 and 5–6. Optional: 3–4 and 7. Complete the required items after instruction by **April 11, 2027, 11:59 p.m. Eastern**. Work individually by hand, check the key, and correct errors. Graded for completion; optional items are excluded from quizzes and the final.

Label variances favorable (F) or unfavorable (U). Items 1 and 2 use different quantities; do not add them as one total usage-cost variance.

1. (*Materials price variance.*) The standard price is \$5.00 per pound; the actual price is \$5.30 per pound; the actual quantity purchased is 8,000 pounds. Compute the direct-materials price variance.
2. (*Materials quantity variance.*) The standard is 3 pounds per unit at \$5.00 per pound; 2,000 units are produced; actual usage is 6,300 pounds. Compute the direct-materials quantity variance.
3. (*Labor rate variance.*) The standard rate is \$20.00 per hour; the actual rate is \$19.50 per hour; actual hours worked total 4,200. Compute the direct-labor rate variance.
4. (*Labor efficiency variance.*) The standard is 2 hours per unit at \$20.00 per hour; 2,000 units are produced; actual hours total 4,200. Compute the direct-labor efficiency variance.
5. (*Return on investment.*) A division reports operating income of \$600,000 and average operating assets of \$3,000,000. Compute ROI.
6. (*Residual income.*) For the same division (operating income \$600,000; operating assets \$3,000,000), the required rate of return is 14%. Compute residual income.
7. (*Economic value added.*) A firm has NOPAT of \$1,200,000, invested capital of \$8,000,000, and a weighted-average cost of capital of 10%. Compute EVA.

Self-Check Answers

Required answers: 1–2 and 5–6. Other answers support optional practice.

1. **Materials price variance.** $= (\text{actual price} - \text{standard price}) \times \text{actual quantity} = (5.30 - 5.00) \times 8,000 = \text{\$2,400 U.}$
2. **Materials quantity variance.** Standard quantity allowed $= 2,000 \times 3 = 6,000$ lbs. Variance $= (\text{actual} - \text{standard}) \times \text{standard price} = (6,300 - 6,000) \times 5.00 = \text{\$1,500 U.}$
3. **Labor rate variance.** $= (\text{actual rate} - \text{standard rate}) \times \text{actual hours} = (19.50 - 20.00) \times 4,200 = \text{\$2,100 F.}$
4. **Labor efficiency variance.** Standard hours allowed $= 2,000 \times 2 = 4,000$. Variance $= (\text{actual} - \text{standard}) \times \text{standard rate} = (4,200 - 4,000) \times 20.00 = \text{\$4,000 U.}$
5. **ROI.** $= 600,000 \div 3,000,000 = \text{20\%}.$
6. **Residual income.** $= \text{operating income} - (\text{required rate} \times \text{operating assets}) = 600,000 - 0.14 \times 3,000,000 = 600,000 - 420,000 = \text{\$180,000}.$
7. **EVA.** $= \text{NOPAT} - (\text{WACC} \times \text{invested capital}) = 1,200,000 - 0.10 \times 8,000,000 = 1,200,000 - 800,000 = \text{\$400,000}.$